QUARANTYNE ACADEMY

Homework Assignment #1

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Class Topic: Security Control Types & Categories

SECTION 1 — MULTIPLE CHOICE (20 Questions | 2 pts each = 40 pts) Circle or clearly write the letter of the best answer.

1. Which control type is designed to STOP a threat before it occurs? A. Detective B. Corrective C. Preventive D. Compensating (C)

2. A warning banner on a login screen is an example of which control type? A. Preventive B. Deterrent C. Detective D. Directive (B)

3. An Intrusion Detection System (IDS) is classified under which control category? A. Physical B. Managerial C. Operational D. Technical (D)

4. After a ransomware attack, your team restores systems from a clean backup. This is a: A. Preventive control B. Deterrent control C. Corrective control D. Compensating control (C)

5. Security patrols conducted by guards fall under which category? A. Technical B. Operational C. Managerial D. Physical (B)

6. A locked server room door is which type AND category combination? A. Detective – Physical B. Preventive – Physical C. Deterrent – Technical D. Directive – Operational (B)

7. An organization cannot patch a legacy system. Instead, they place it on an isolated VLAN. This is: A. Corrective – Technical B. Preventive – Technical C. Compensating – Technical D. Detective – Managerial (C)

8. Which control type guides behavior through rules, policies, and procedures? A. Deterrent B. Compensating C. Directive D. Preventive (C)

9. A smoke detector that alerts staff to a fire is an example of which control type? A. Corrective B. Preventive C. Deterrent D. Detective (D)

10. The Employee Handbook falls under which category? A. Technical B. Physical C. Operational D. Managerial (C)

11. Which of the following is a MANAGERIAL Preventive control? A. Firewall B. Locked Door C. Security Policy D. Security Patrol (C)

12. A CCTV sign posted at a building entrance is which type of control? A. Detective – Physical B. Deterrent – Physical C. Preventive – Physical D. Directive – Physical (B)

13. Risk Acceptance documentation is an example of which combination? A. Compensating – Managerial B. Corrective – Operational C. Detective – Technical D. Directive – Managerial (A)

14. Multi-factor authentication (MFA) is best classified as: A. Detective – Technical B. Preventive – Technical C. Corrective – Operational D. Deterrent – Physical (B)

15. Which control type is used when the ideal control cannot be implemented? A. Deterrent B. Directive C. Compensating D. Corrective (C)

16. A compliance requirement such as FISMA is which type and category? A. Directive – Managerial B. Preventive – Technical C. Detective – Operational D. Compensating – Physical (A)

17. An antivirus program that quarantines a malicious file is: A. Preventive – Technical B. Detective – Technical C. Corrective – Technical D. Deterrent – Technical (C)

18. Which of the following best represents a PHYSICAL Detective control? A. Locked Door B. Motion Sensor C. Safety Sign D. Emergency Exit (B)

19. Dual Authorization (requiring two people to approve an action) is an example of: A. Corrective – Technical B. Compensating – Operational C. Preventive – Managerial D. Detective – Physical (B)

20. A password policy enforced by the system is which combination? A. Directive – Technical B. Preventive – Managerial C. Detective – Operational D. Corrective – Technical (A)

SECTION 2 — THEORY QUESTIONS (20 Questions | 3 pts each = 60 pts) Answer in complete sentences. Where asked for examples, use your own — do not copy from class slides.

1. In your own words, define the term 'security control.' Why are security controls important in an organization?

Answer: A security control is any safeguard or measure put in place to protect systems, data, and people from threats. They are important because they help prevent incidents, detect issues early, and ensure the organization can respond and recover effectively.

2. Explain the difference between a Preventive and a Deterrent control. Give one original example of each.

Answer: Preventive controls stop an issue from happening, while deterrent controls discourage someone from attempting it.

- Preventive example: Access request approval workflows (ServiceNow). This is Operational.
- Deterrent example: Account lockout warnings. This is Technical.

3. What is a Detective control? Describe a real-world scenario where a detective control would be critical.

Answer: A detective control identifies when something has already happened or is happening.
Example: In a bank, transaction monitoring systems detect suspicious activity like fraud in real time**.**

4. Explain what a Corrective control does and at what point in the security lifecycle it is applied.

Answer: Corrective controls fix problems after they are detected. They are applied after an incident occurs.
Example: Restoring systems after a cyberattack.

5. What is a Compensating control? Describe a situation where an organization would need to use one.

Answer: A compensating control is used when the ideal control cannot be implemented.
Example: If a system cannot support MFA, the organization may increase monitoring and logging.

6. Define the Directive control type and explain why policies and procedures are considered controls.

Answer: Directive controls guide behavior using policies and rules. They are considered controls because they set expectations and influence how people act.

7. Compare and contrast Technical and Physical control categories. Give two examples of each.

Answer:

- Technical controls involve systems (firewalls, MFA)
- Physical controls protect physical access (locks, guards)

8. What is the difference between a Managerial control and an Operational control? Why does the distinction matter?

Answer: Managerial controls are policies and governance, while operational controls are day-to-day procedures.
It matters because both strategy and execution are needed for security to work.

9. Can a single control belong to more than one control type? Explain your answer with an example.

Answer: Yes. One control can serve multiple purposes.
Example: A locked door prevents access and deters intruders.

10. Using the Controls Matrix (Category + Type), explain how a 'Log Review Mandate' fits into the matrix and why it is placed there.

Answer: It is Detective – Managerial because it is a policy requiring logs to be reviewed to detect issues.

11. A small business has no budget for new technology. What types of controls can they implement using only people and processes? Explain using control categories.

Answer: They can implement:

- Managerial: policies
- Operational: procedures and training
- Physical: locks and signs

12. Why is an Emergency Exit classified as a Corrective – Physical control rather than a Preventive control?

Answer: It is Corrective – Physical because it helps respond to an incident, not prevent it.

13. Explain the role of the Managerial category in a security program. Why can't an organization rely solely on Technical controls?

Answer: Managerial controls define strategy and governance. Organizations cannot rely only on technical controls because people and processes are equally important.

14. In the context of federal cybersecurity (e.g., FISMA, NIST RMF), which control type do compliance requirements like FISMA represent and why?

Answer: It is Directive – Managerial because it sets compliance requirements organizations must follow.

15. Describe a scenario in a hospital setting and identify at least three different control types that would be applied. Explain each one.

Answer:

- Preventive: MFA for patient records
- Detective: Log monitoring for access
- Corrective: Incident response after breach

16. What is the relationship between a Security Policy (Preventive – Managerial) and an Acceptable Use Policy (Deterrent – Managerial)? How do they work together?

Answer: Security Policy defines rules, while AUP reinforces acceptable behavior. Together, they guide and enforce security expectations.

17. Explain why Security Awareness Training is classified as a Preventive – Operational control and not a Technical one.

Answer: It is Preventive – Operational because it trains people to avoid mistakes, not a system-based control.

18. A new employee at your organization asks: 'Why do we have so many rules and signs posted everywhere?' How would you explain directive controls to them in simple terms?

Answer: Directive controls are rules and guidelines that help everyone know what is expected so we can avoid mistakes and security issues.

19. Describe a scenario where a Compensating control was used in your personal life (home, school, or work). Identify its category.

Answer: At home, if I don't have a smart security system, I double-check locks and use cameras. That's a Physical/Operational compensating control.

20. If you were designing a security program for Quarantyne Academy, which three controls would you prioritize first and why? Identify each by type and category.

Answer:

- MFA (Preventive – Technical)
- Security Policy (Preventive – Managerial)
- Incident Response Plan (Corrective – Managerial)
  These provide prevention, governance, and recovery.

SECTION 3 — LABS (4 Labs | 10 pts each = 40 pts) Complete each lab exercise. Document your steps, reasoning, and results clearly.

LAB 1 — Control Classification Exercise (10 pts) Below is a list of 10 security controls. For each one, identify: (a) the Control Type and (b) the Control Category. Present your answers in a table.

Create a 3-column table: Control Name | Control Type | Control Category

Control	Type	Category
Firewall	Preventive	Technical
Security guard at building entrance	Deterrent	Physical
SIEM log monitoring	Detective	Technical
Incident Response Plan	Corrective	Managerial
Biometric door lock	Preventive	Physical
Password complexity policy	Directive	Technical
Risk Acceptance memo signed by leadership	Compensating	Managerial
Employee badge access system	Preventive	Physical
Security awareness training for new hires	Preventive	Operational
Backup restoration after data loss	Corrective	Operational

LAB 2 — Build a Mini Controls Matrix (10 pts)

For a fictional small business called 'Bright Eats Catering,' create a Controls Matrix with all 6 control types and all 4 categories (Technical, Managerial, Operational, Physical). Fill in at least one control for each cell. You may use controls from class or invent realistic ones.

Your matrix must be a 6x4 grid. You may draw it, type it, or create it in a spreadsheet and attach a screenshot.

Type/Category	Technical	Managerial	Operational	Physical
Preventive	MFA	Security Policy	Staff Training	Locked Storage
Deterrent	Warning Banner	AUP	Policy Posters	CCTV Sign
Detective	SIEM	Audit Policy	Log Reviews	Motion Sensor
Corrective	Antivirus	IR Plan	Backup Restore	Fire Exit
Compensating	VLAN	Risk Acceptance	Dual Approval	Guard
Directive	Password Rules	Compliance Policy	SOP	Safety Signs

LAB 3 — Scenario Analysis (10 pts)

Read the following scenario and answer the questions below. Scenario: A hospital stores patient records on a server. Last week, an unauthorized employee accessed confidential files. The IT team discovered the breach through log alerts three days later. Management is now implementing new controls.

• a. What type of control detected the breach? What category does it fall under?

Answer: Detective – Technical (Log alerts)

• b. What Preventive controls could have stopped the breach before it happened?

Answer:

- MFA
- Role-based access control
- Data access restrictions

• c. What Corrective controls should be applied now?

Answer:

- Reset access credentials
- Restore compromised data
- Incident response execution

• d. What Directive control would you recommend to prevent future incidents?

Answer: Data access policy or acceptable use policy

• e. Is there a need for a Compensating control here? Explain why or why not.

Answer: Yes — if system limitations exist, increased monitoring or manual approvals can be used.

LAB 4 — True or False + Justify (10 pts)

State whether each statement is TRUE or FALSE, then write 1–2 sentences justifying your answer.

• 1. A locked door can serve as both Preventive and Deterrent control.

Answer: True. It prevents access and deters behavior.

• 2. Managerial controls always involve technology.

Answer: False. Managerial controls are policies, not technology

• 3. A Compensating control is only used when no other option exists.

Answer: False. Compensating controls are used when ideal control is not feasible, not only last resort

• 4. Security patrols are an example of a Physical control.

Answer: False. Security patrols are Operational, not purely Physical

• 5. Directive controls guide behavior but cannot prevent a security incident on their own.

Answer: True. They guide behavior but don't enforce directly

SECTION 4 — PROJECTS (4 Projects | 10 pts each = 40 pts)

Each project must be submitted as part of your PDF. Upload your completed homework PDF to your GitHub repository as a project. Be prepared to present and defend your work in the next class.

PROJECT 1 — Personal Security Audit (10 pts)

Conduct a mini security audit of your home or personal digital life. Identify 6 security controls you currently have in place (or are missing). For each, identify the control type and category. Present your findings in a structured write-up with a summary table.

- Your write-up must be at least one full page.
- Include at least one control from each category (Technical, Physical, Operational, Managerial).
- Recommend at least two additional controls you should implement and explain why.

Control	Type	Category
MFA on accounts	Preventive	Technical
Home door locks	Preventive	Physical
Security awareness habits	Preventive	Operational
Password policy (personal)	Directive	Managerial
Phone tracking enabled	Detective	Technical
Camera at home	Detective	Physical

I currently use a mix of technical, physical, operational, and managerial controls in my personal life. These controls help protect my data, home, and devices. However, I recognize gaps such as lack of full home automation security and centralized monitoring.

Recommendations:

1. Install a smart home security system (Preventive – Technical)

2. Enable backup and recovery automation (Recovery – Technical)

PROJECT 2 — Security Controls Poster / Infographic (10 pts)

Design a one-page visual reference guide (infographic or poster) that explains all 6 Security Control Types and all 4 Categories. Your poster must include: a brief definition of each type, one real-world example per type, and a visual that makes the content easy to understand for a new student.

- You may use Canva, PowerPoint, Google Slides, or any design tool.
- Export as an image or PDF and include it in your submission.
- It must clearly reflect Quarantyne Academy's black and gold brand colors.

QUARANTYNE ACADEMY

Security Controls: Types & Categories

A VISUAL REFERENCE GUIDE FOR CYBERSECURITY STUDENTS

6 SECURITY CONTROL TYPES

Preventive

Stops a threat or attack from occurring in the first place.

EXAMPLE
Firewall blocking unauthorized inbound traffic

Detective

Identifies and alerts on security incidents that have already occurred.

EXAMPLE
IDS alerting on unusual network traffic patterns

Corrective

Reduces the impact of an incident and restores systems to normal.

EXAMPLE
Restoring files from backup after ransomware attack

Deterrent

Discourages attackers by making an attack seem costly or risky.

EXAMPLE
Security camera signs and warning banners on login pages

Compensating

Alternative control used when a primary control isn't feasible.

EXAMPLE
MFA as a substitute when single sign-on isn't available

Directive

Guides behavior through policies, rules, and procedures.

EXAMPLE
Acceptable use policy requiring strong passwords

4 CONTROL CATEGORIES

Technical

Hardware & software controls — firewalls, encryption, access controls

Physical

Real-world barriers — locks, guards, fences, security cameras

Administrative

Policies & procedures — security training, hiring checks, audits

Operational

Day-to-day practices — patch management, incident response, backups

TYPES × CATEGORIES — QUICK REFERENCE MATRIX

Control Type	Technical	Physical	Administrative	Operational
Preventive	Firewall, ACL	Door lock, fence	Hiring policy	Patch management
Detective	IDS / SIEM	Motion sensor	Audit log review	Security monitoring
Corrective	Antivirus quarantine	Sprinkler system	Incident response plan	Backup restoration
Deterrent	Login warning banner	CCTV signage	Disciplinary policy	Security awareness
Compensating	MFA, encryption	Security escort	Exception policy	Manual review process
Directive	Config standards	Signage, rules	Acceptable use policy	SOPs, checklists

PROJECT 3 — Organizational Security Control Plan (10 pts)

You have been hired as a GRC analyst for a fictional startup called 'SwiftPay' — a mobile payment app company. Create a basic Security Control Plan that includes at minimum 12 controls across all 6 types and all 4 categories. For each control, document: the control name, type, category, and a brief description of how it will be implemented.

- Present your plan in a table or structured document format.
- Explain in a short paragraph (3–5 sentences) why you prioritized the controls you chose.
- Identify one gap area and recommend a compensating control for it.

Security Control Plan (SwiftPay – Mobile Payment App)

Control Name	Type	Category	Description
Multi-Factor Authentication (MFA)	Preventive	Technical	Enforce MFA for all user and admin logins to prevent unauthorized access
Web Application Firewall (WAF)	Preventive	Technical	Protect APIs and web traffic from malicious attacks
Security Policy	Preventive	Managerial	Define rules for data protection, access, and usage across the organization
Acceptable Use Policy (AUP)	Deterrent	Managerial	Inform users of acceptable system usage and consequences of misuse
SIEM Monitoring (Splunk)	Detective	Technical	Monitor logs and detect suspicious activity in real time
Internal Audit Reviews	Detective	Managerial	Conduct periodic audits to ensure compliance with policies and controls

Control Name	Type	Category	Description
Incident Response Plan	Corrective	Managerial	Define steps to respond to and recover from security incidents
Backup & Recovery Process	Corrective	Operational	Regularly back up data and restore systems after incidents
Network Segmentation (VLAN)	Compensating	Technical	Isolate sensitive systems when full security controls are not possible
Risk Acceptance Documentation	Compensating	Managerial	Document and approve risks that cannot be mitigated immediately
Standard Operating Procedures (SOPs)	Directive	Operational	Provide step-by-step guidance for employees on secure processes
Security Awareness Training	Preventive	Operational	Train employees to recognize and avoid security threats
Badge Access Control	Preventive	Physical	Restrict physical access to sensitive areas
CCTV Monitoring	Detective	Physical	Monitor physical spaces for suspicious activity

PROJECT 4 — Reflection Essay (10 pts)

- Write a 1–2 page reflection essay answering the following questions:
- Why did you join Quarantyne Academy and what do you hope to achieve?
- Before this class, what did you know (or think you knew) about cybersecurity controls?
- What surprised you most about the 6 control types or the controls matrix?
- How do you plan to apply what you learned today in your career or personal life?
- What questions do you still have that you hope will be answered in future classes?

I joined Quarantyne Academy because I wanted to strengthen my understanding of cybersecurity concepts and align them with my experience as a Business Analyst, especially in industries like aviation, banking, and healthcare. My goal is to not only understand security from a theoretical standpoint but also be able to apply it practically when gathering requirements, working with stakeholders, and supporting system implementations. I see cybersecurity as a critical part of digital transformation, and I want to position myself as a BA who can bridge both business and security needs.

Before this class, I had a general understanding of security controls, mostly from a functional perspective-things like access control, authentication, and compliance requirements. However, I did not fully understand how controls are categorized into types like preventive, detective, corrective, deterrent, compensating, and directive. I also hadn't thought deeply about how these controls are further broken down into categories such as technical, managerial, operational, and physical. This class helped me connect those concepts in a more structured and meaningful way.

What surprised me the most was how a single control can serve multiple purposes depending on how it is implemented. For example, something as simple as a locked door can be both preventive and deterrent. The controls matrix gave me a clearer framework for thinking about security holistically, rather than just focusing on one aspect like technology. It showed me that effective security requires a combination of people, processes, and technology working together.

I plan to apply what I've learned in my career, especially in ServiceNow projects where I help define requirements, workflows, and controls for systems like ITSM, CSM, and ITOM. Understanding control types will help me ask better questions during requirement gathering, ensure compliance is built into solutions, and support risk management efforts more effectively. On a personal level, I can also apply these concepts to improve my own digital security practices, such as strengthening authentication and being more aware of potential risks.

Going forward, I would like to learn more about how these controls map to real-world frameworks like NIST, ISO, and FISMA, and how organizations implement and audit them in practice. I am also interested in understanding how tools like ServiceNow GRC automate control management and compliance tracking. Overall, this course has given me a strong foundation, and I am looking forward to building on it.

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